SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 7 – Perturbations

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Second Tutoring session

Friday 24 October

Room L.11

14:15 – 16:15

Due to Wednesday 22nd - Write questions to me: annachiara.ippolito@polimi.it

These hours are meant to fill the gap on the fundamentals of MATLAB and Simulink.

Partecipate if you have questions: it will not be a lecture but an interactive session!

www.tutorapp.polimi.it -> try doing a first access!

The day of the tutoring you must register **within the first half an hour.**

My person code: 10661425

Task 1: Include Keplerian dynamics in your model

Tasks

- Augment your model by adding **two-body Keplerian dynamics.** n is the average rotational rate of the orbit. <u>You choose</u> the orbital parameters.

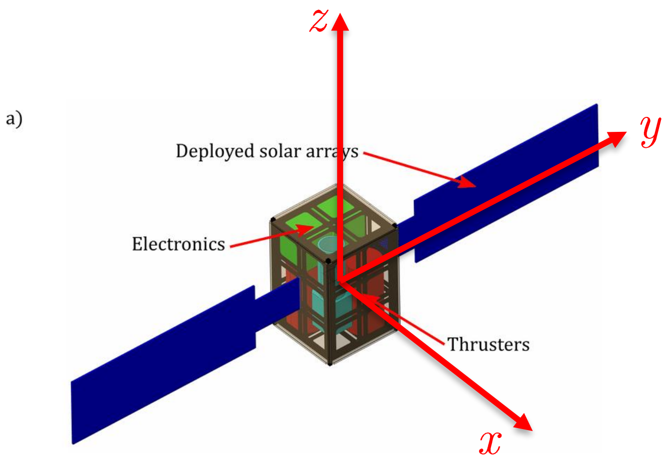
$$\mathbf{r}_N = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = r \begin{bmatrix} \cos\theta \\ \sin\theta\cos i \\ \sin\theta\sin i \end{bmatrix} \qquad r = \frac{a(1-e^2)}{1+e\cos\theta} \qquad \dot{\theta} = \frac{n(1+e\cos\theta)^2}{(1-e^2)^{3/2}}$$

- Compute $\mathbf{r}_B$ exploiting the DCM.

- Augment your model by computing the **Sun direction** in the **inertial and body frame**.

$$\mathbf{S}_N = r_\odot \begin{bmatrix} \cos(n_\odot t) \\ \sin(n_\odot t)\cos\varepsilon \\ \sin(n_\odot t)\sin\varepsilon \end{bmatrix} \qquad n_\odot = 2\pi/T \qquad T = 1 \text{ year} \qquad \varepsilon = 23.45 \text{ deg}$$

Task 2: Model the Solar Radiation Pressure (SRP) torque



$$J_{\mathrm{depl}} = \begin{bmatrix} 100.9 & 0 & 0 \\ 0 & 25.1 & 0 \\ 0 & 0 & 91.6 \end{bmatrix} \cdot 10^{-2} \; [\mathrm{kg\ m^2}]$$

When calculating the torque, what happens if the face is not illuminated? **You have to consider it!**

Tasks

- Model the **SRP torque**. The SRP force for each of the satellite exposed face is given by the following expressions:

$$\mathbf{F}_i = -PA_i\left(\hat{\mathbf{S}}_B \cdot \hat{\mathbf{N}}_{Bi}\right)\left[(1-\rho_s)\hat{\mathbf{S}}_B + \left(2\rho_s\left(\hat{\mathbf{S}}_B \cdot \hat{\mathbf{N}}_{Bi}\right) + \frac{2}{3}\rho_d\right)\hat{\mathbf{N}}_{Bi}\right] \qquad P = \frac{F_e}{c}$$

with c being the speed of light and F_e the solar radiation intensity.
Assume the latter constant at $1358\ \mathrm{W/m^2}$

- Assume the following properties for the satellite exposed surfaces. **Calculate the torque** (with respect to the center of mass). Then, try to apply a **fixed shift of the C.M.** with respect to the geometric center.

	$\hat{N}_B$	ρ_s	ρ_d	A [m2]	$r_{F,i}$ [cm]
Body1	[+1,0,0]	0,5	0,1	$6\cdot10^{-2}$	[+10,0,0]
Body2	[0,+1,0]	0,5	0,1	$6\cdot10^{-2}$	[0,+10,0]
Body3	[-1,0,0]	0,5	0,1	$6\cdot10^{-2}$	[-10,0,0]
Body4	???	0,5	0,1	$6\cdot10^{-2}$	???
Body5	???	0,5	0,1	$4\cdot10^{-2}$	???
Body6	[0,0,-1]	0,5	0,1	$4\cdot10^{-2}$	[0,0,-15]
Panel1	[+1,0,0]	0,1	0,1	$12\cdot10^{-2}$	[0,45,0]
Panel2	[-1,0,0]	0,1	0,1	$12\cdot10^{-2}$	[0,+45,0]
Panel3	[+1,0,0]	0,1	0,1	$12\cdot10^{-2}$	[0,-45,0]
Panel4	???	0,1	0,1	$12\cdot10^{-2}$	[0,-45,0]

Task 3: Add the Gravity Gradient for inclined and elliptical orbit

Tasks

- Add the **gravity gradient perturbation**.

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x}\omega_y\omega_z \boxed{- \frac{3GM_t}{r^3}\frac{I_y - I_z}{I_x}c_2c_3} \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y}\omega_x\omega_z \boxed{- \frac{3GM_t}{r^3}\frac{I_z - I_x}{I_y}c_1c_3} \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z}\omega_y\omega_x \boxed{- \frac{3GM_t}{r^3}\frac{I_x - I_y}{I_z}c_2c_1} \end{cases}$$

with

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = A_{B/L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

When adding the gravity gradient, you Must consider the non-circularity of your new orbit. The new matrix is

$$A_{L/N} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix}$$

Task 4: Add the magnetic torque due to Earth's magnetic field.

Tasks

- Compute the **magnetic field vector** in the inertial reference frame.
  Use the DGRF model, in which magnetic field is computed as the gradient of the scalar potential

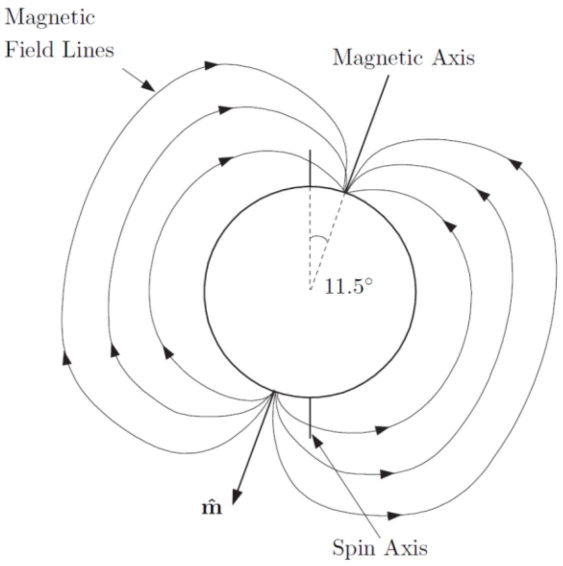
$$V(r,\phi,\theta) = R\sum_{n=1}^{k}\left(\frac{R}{r}\right)^{n+1}\sum_{m=0}^{n}\left[g_n^m\cos(m\phi)+h_n^m\sin(m\phi)\right]P_n^m\left(\cos\theta\right)$$

  The coefficients can be found in the table to the right, giving a value in nT. On a first approximation, you can assume an expression with n=1:

$$\mathbf{b}_N = \frac{R_\oplus^3 H_0}{r^3}[3(\hat{\mathbf{m}}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}}-\hat{\mathbf{m}}] \qquad H_0=\sqrt{\left(g_1^0\right)^2+\left(g_1^1\right)^2+\left(h_1^1\right)^2} \qquad \hat{\mathbf{m}} = \begin{bmatrix}\sin(11.5^\circ)\cos(\omega_\oplus t)\\ \sin(11.5^\circ)\sin(\omega_\oplus t)\\ \cos(11.5^\circ)\end{bmatrix}$$

- Convert the magnetic field vector in body frame.

- The torque acting on the spacecraft can be computed as
  $\mathbf{M}=\mathbf{j}_B\times\mathbf{b}_B \qquad \mathbf{j}_B=[0.01\quad 0.05\quad 0.01]^{\mathrm{T}}\ \mathrm{A}\cdot\mathrm{m}^2$
  in this expression, the spacecraft act as a coil with flowing current.



		DGRF 2020	
n	m	g_n^m	h_n^m
1	0	-29404,8	-
1	1	-1450,9	4652,5
2	0	-2499,6	-
2	1	2982,0	-2991,6
2	2	1677,0	-734,6
3	0	1363,2	-
3	1	-2381,2	-82,1
3	2	1236,2	241,9
3	3	525,7	-543,4

(Bonus) Task 5: Plot the disturbances

Tasks

Plot all the disturbances modelled in the previous task.

- Which one is the strongest?
- What happens if you change your orbit parameters?
- Reason on your results.

- *…are you really plotting that on a linear scale?*